Enrolment No.

NITM/XXXXXX

NATIONAL INSTITUTE OF TECHNOLOGY, MANIPUR

B.Tech. Semester – VI (CSE), Mid Term Examination – 2025

Subject: Compilers (CS 304)

Total Marks: 30 Time: $1\frac{1}{2}$hours

The figures in the margin indicate full marks for the questions

All Questions are compulsory.

1.	a) Given a grammar G with productions: S→ACB | CbB | Ba, A→d*a* | BC, B→g | ε, C→h | ε, find the first and follow of the non-terminals and construct the top-down predictive parsing table.	[5]
	b) Given a grammar G with productions S→bSSaaS | bSSaSb | bSb | a, write the grammar after left factoring G.	[3]
	c) Explain the difference between compiler and interpreter with examples.	[2]
2.	a) Construct an NFA from a regular expression $(ab^* \mid a^*)^*$ using Thompson algorithm and convert the NFA to a DFA using subset construction.	[5]
	b) Using the grammar G and predictive parsing table of Question 1.a), check whether the string ***ghhg*** is accepted by the grammar using Top Down Predictive Parsing.	[3]
	c) Explain the structure and the functions of a Lexical Analyzer which include simulating an automaton with a diagram.	[2]
3.	a) Write a lex program to count the number of words starting with a, b and c from the input strings.	[4]
	b) Given a left recursive grammar G with productions S→A, A→aB | Ad, B→b, C→g, transform the grammar G into non- left recursive grammar.	[3]
	c) Given a grammar G with productions S→SS+ | SS* | a. Check whether the grammar G is ambiguous or not using the string aa+a*.	[3]

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